University of Cape Town



Machine Learning Assignment 1

Pamela Afful

Student Number: AFFPAM001

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Department of Statistical Sciences Plagiarism Declaration form

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COURSE CODE: **STA** 5068Z

COURSE NAME: MACHINE LEARNING

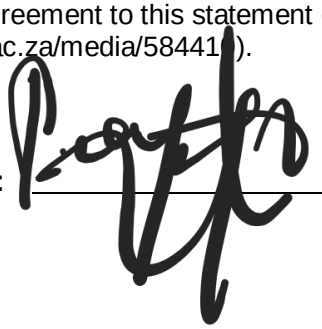
STUDENT NAME: PAMELA AFFUL

STUDENT NUMBER: AFFPAM001

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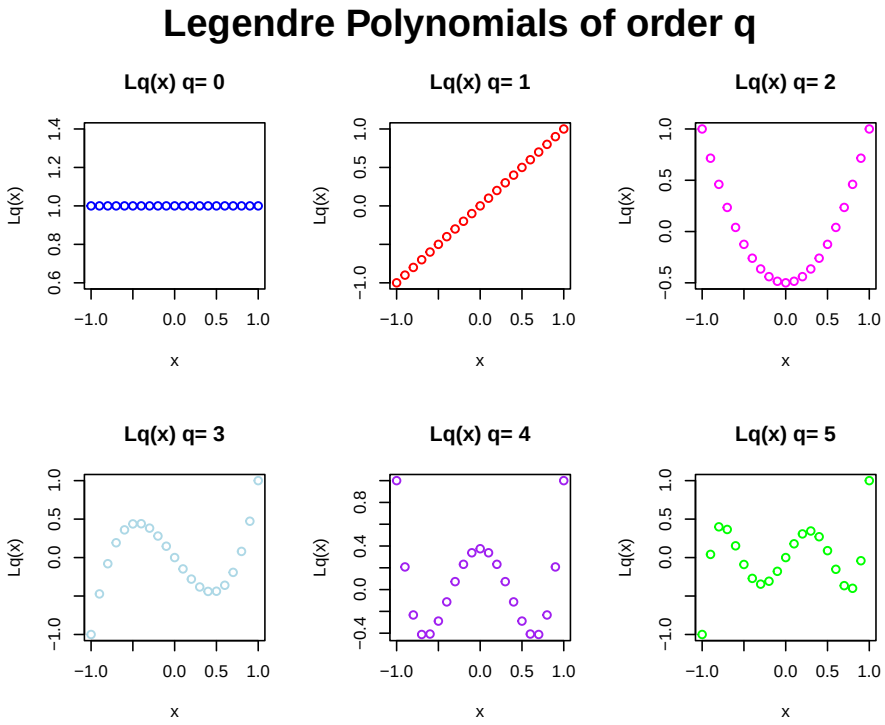
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Question 1

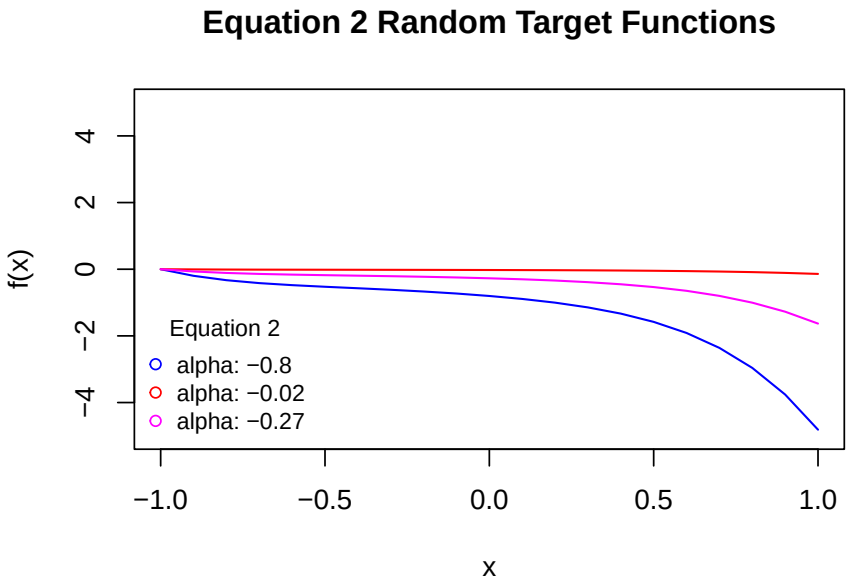
a) Legendre Polynomial Plots

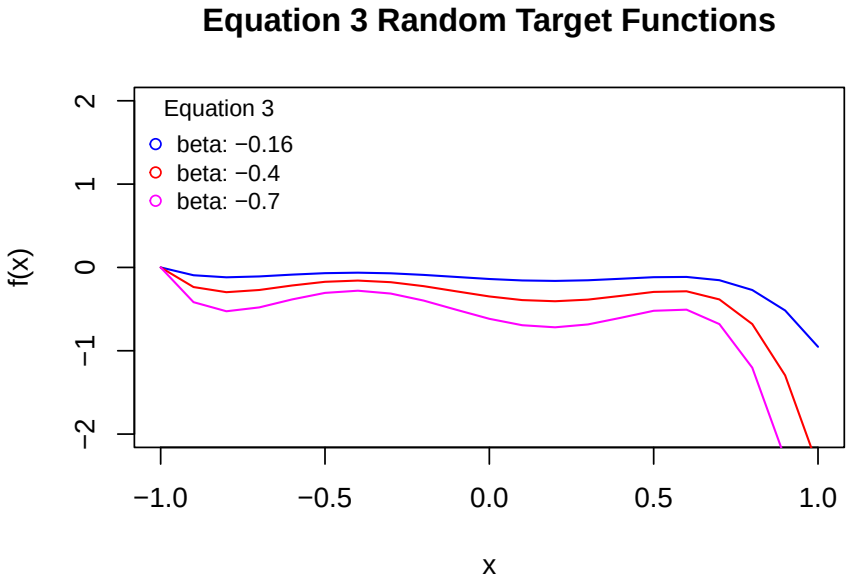


As the order of the polynomial, q, increases, the more nonlinear/complex the $L_q(x)$ becomes:

1. At q=0, $L_q(x)$ devolves to a straight horizontal line $L_q(x)$ =1 over the domain [-1,1].
2. At q=1, $L_q(x)$ starts to have some slope attached, a straight line of the form $y = 2x$.
3. At q=2 the non-linear curvature/complexity of $L_q(x)$ starts to become more visible, particularly towards the edges of the domain [-1,1]; the function approximates a parabola/quadratic function.
4. At q=3,4,5, the non-linear, polynomial curvature/complexity of $L_q(x)$ becomes more prominant with the function taking more sinusoidal shape.

b) Random Target Functions



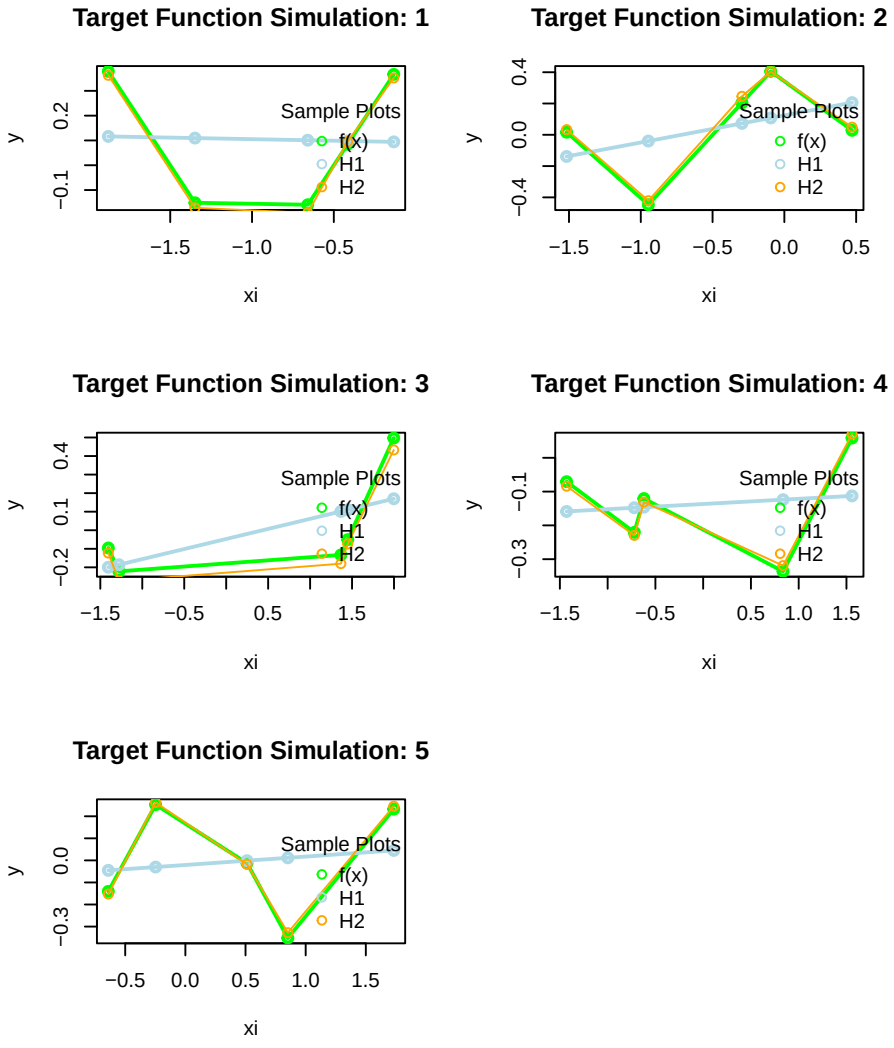


Observations:

1. $f(x) = \sum_{q=0}^{Q_f} \alpha_q x^q$: the functions only output values from 0 to negative values over the domain [-1,1], although the lines are smoother with very little curvature in between.

2. $f(x) = \sum_{q=0}^{Q_f} \beta_q L_q(x)$: the functions also output 0 to negative values over the domain; however, over the domain [-1,1], however, for certain values of β there are curvatures, with the function taking a wave-like shape.

Question 2



b) A Priori expectation

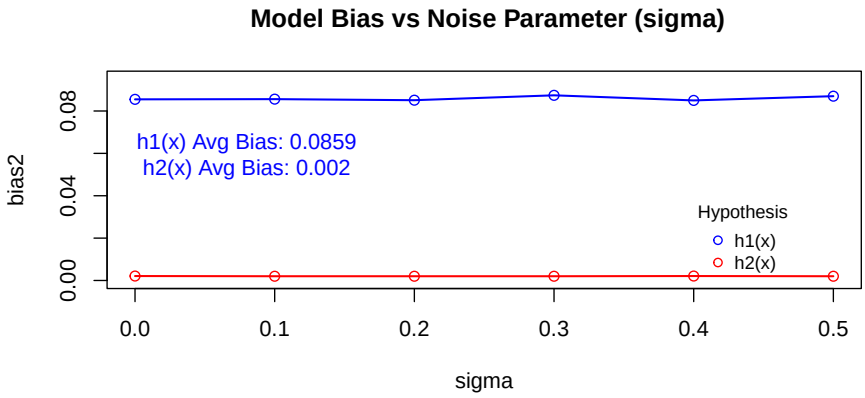
Given that the underlying data generating process is non-linear, the non-linear hypothesis would be more suited to fit the data's non-linear behavior and overall shape. This visually seems to be the case as well: the H2(x) hypothesis functions follows the shape of the data quite closely. The linear function only fits a straight line through the data with only a few values intersecting with the underlying

function output values.

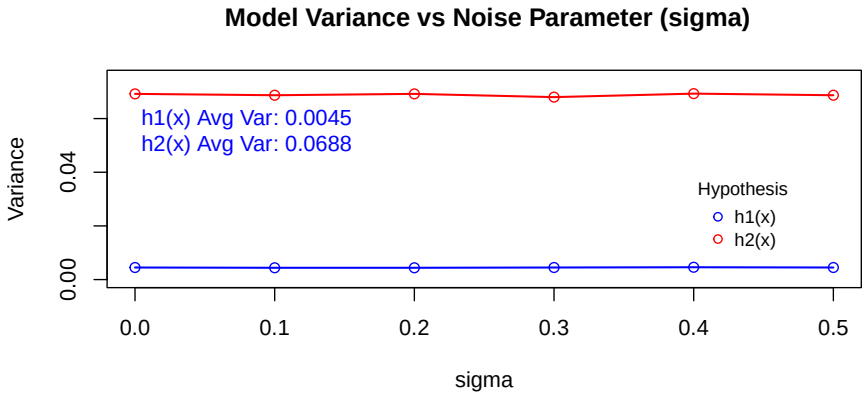
c) Bias Variance for H1 and H2

	Bias	Variance	Both:Bias+Var	OOS Error	Estimate Differance
h1(x)	0.0854	0.0044	0.0898	0.0841	0.0057
h2(x)	0.0021	0.0709	0.0730	0.1295	0.0565

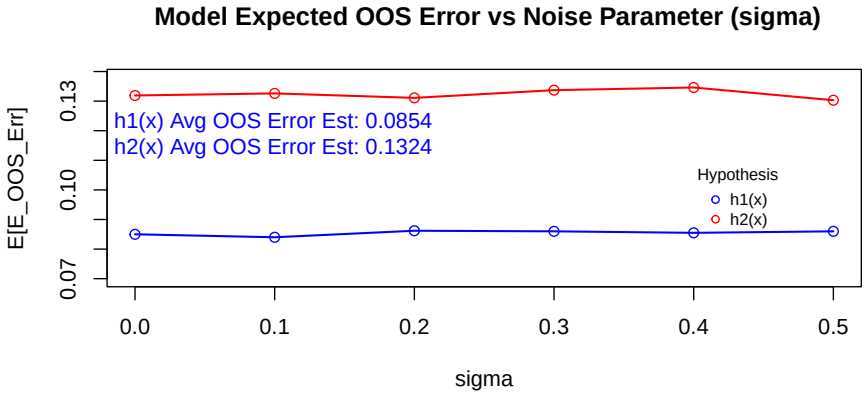
d) Bias Variance Dynamics



(a) Bias vs Noise



(b) Variance vs Noise



(c) E[OOS Error] vs Noise

Figure 1: OOS Error Estimates

e) Interpretation

Bias-Variance Dynamics

The expected out-of-sample error as it relates to the bias and variance of a model is given by

$$E[Err_{oos}] = Bias^2 + Variance + IrreducibleNoise(\varepsilon)$$

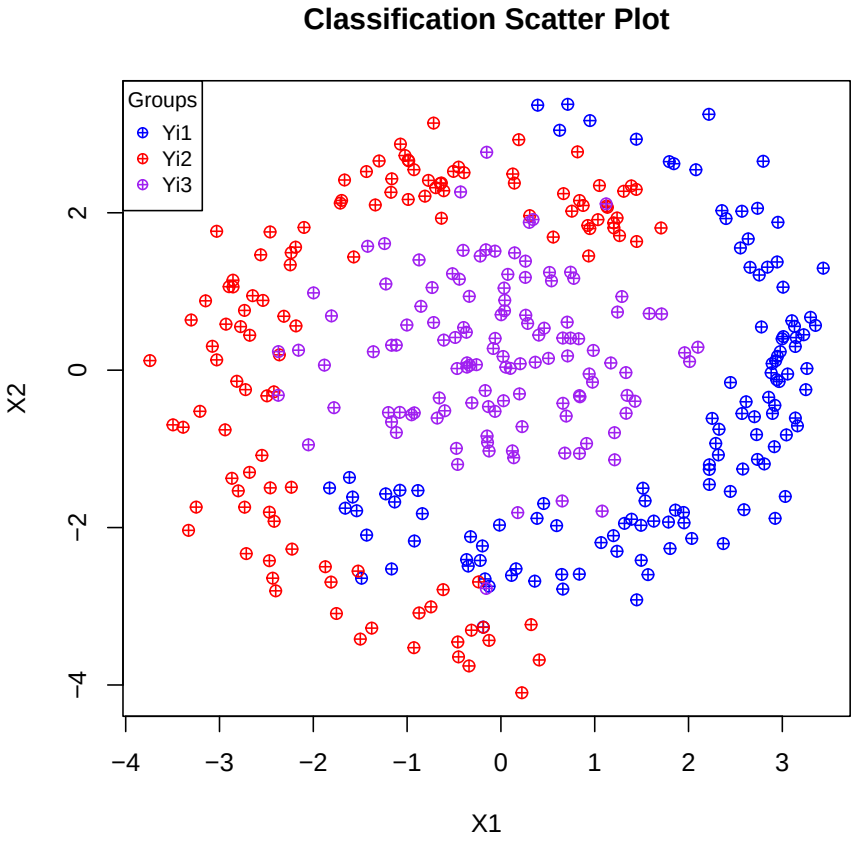
- RHS:
  - The linear model, $h1(x)$, has better out-of-sample performance overall, averaged across 1000 simulations over varying values of the noise parameter, compared to the more complex model, $h2(x)$.
  - There are relatively small changes in the average OOS error for both hypothesis functions. The steepest increase in the average OOS error is observed in the change of the noise parameter from 0.1 to 0.2 and levels thereafter for $h1(x)$ and mostly increases for $h2(x)$ as the noise parameter increases from 0.2.
- LHS:
  - There is very little variability/ change observed in the variance as the noise parameter increases for both functions, whilst the bias for $h1(x)$, sees a small decrease when the noise level increases from 0.3 to 0.4.
  - Overall, the linear model, $h1(x)$, has lower average variance but higher bias across the various noise parameters; this is expected given the that the model is less complex relative to $h2(x)$.

Model Choice

- Given the estimated OOS Error, it's better to go with the linear model as it is more parsimonious and more likely to be able to generalize to new data as observed across the above simulations.
- Rational: the combined estimate comparison $bias^2 + Var$ to $E[Err_{oos}]$ across sigmas :
  - $h1(x)$: its average RHS, 0.0904, is closer to its LHS OOS estimate, 0.0854, providing a better estimate of the model's general OOS behavior compared to $h2 = (x)$; its average RHS, 0.0708 has a much larger deviation from its average LHS estimate, 0.1323.

Question 4

a) Data Set Plot



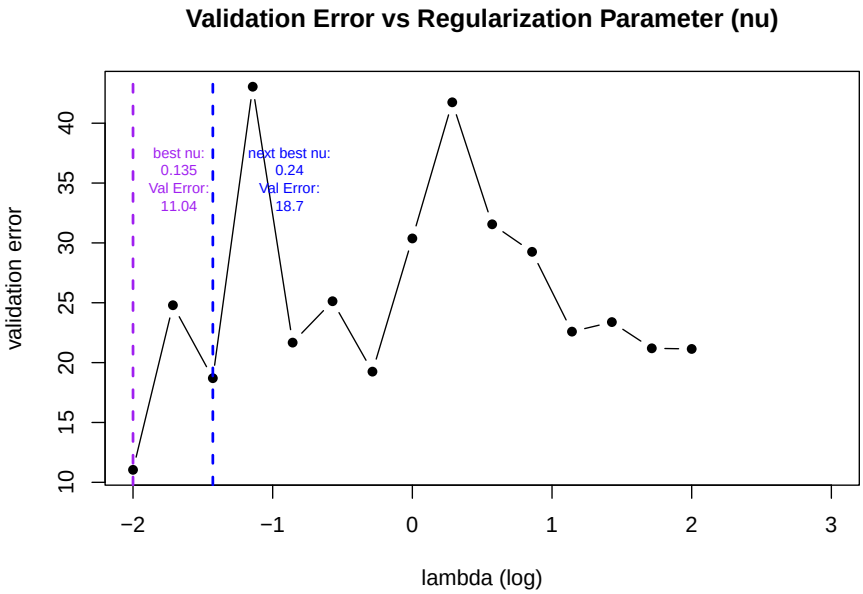
Observation: The data does not appear to be linearly separable (in the given 2D feature space) as the data does not seem to have linear boundaries between categories/classes. Instead, the different classes of responses look to be spiraled around each other. A non-linear neural network seems well suited to capture the non-linear pattern/structure embedded in the data for the given feature space.

c) Feed Forward Neural Network

Numerical Pitfalls: The computation of $log(\hat{y})$ in the objective function may cause the network to explode/ diverge given the undefined behavior of the log function at 0 (or values very close to 0), which can hamper the numerical stability of the network's computations.

d) Validation Analysis

Plot the validation error vs log (λ)

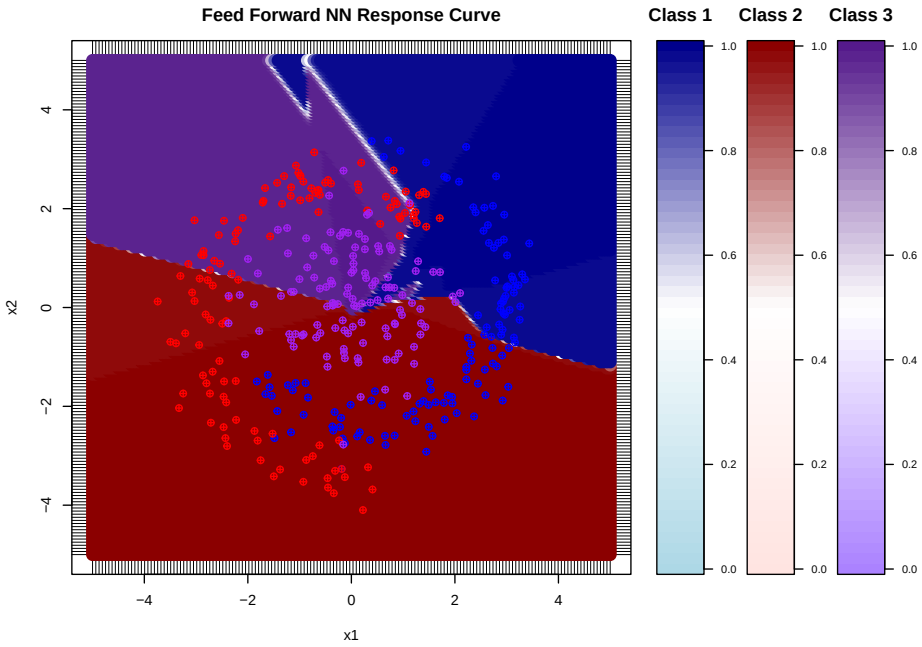


Model Parameters: number of hidden layer nodes, **m =10**

The lowest validation error is generated at $\lambda_{min} = 0.135$ with an associated validation error of 11.04, making for a less strict and more complex model (given the big budget that the sum of the model's parameters cannot exceed). The next best error to the right of this minimum is obtained at $\lambda_{1SE} = 0.24 (Val_{err} = 18.7)$, offering stricter control over the complexity of the network and thus more parsimonious model, decreasing the likelihood of the model over fitting to just the training data. **This is the better choice of regularization parameter.**

d) Response Curve

Model Parameters: **m=10, nu=0.24**



The model produces three distinct, somewhat jagged regions over the feature space, each corresponding to the probability assignments for one class:

- Class 1: Roughly over half of the Class 1 observations are misclassified. The upper-right region captures only some of them, while a cluster near the bottom (X1 between about -2 and 3, X2 between about -3 and -1.5) falls within the Class 2 region. The model is very confident within its Class 1 region, assigning probabilities close to 1, so new observations resembling that bottom cluster would likely be misclassified as Class 2.

- Class 2: Class 2 predictions dominate the feature space. The region lies below a boundary that slopes from X2 $\sim$ 1.4 at (X1 = -5) down to $\sim$ -1.2 (at X1 = 5), and the model predicts Class 2 there with probabilities close to 1. However, a significant portion of Class 2 observations (X1 between about -3 and 1, X2 between about 1.5 and 3) falls within the Class 3 region.

- Class 3: The Class 3 region extends from the left edge (X1 = -5) to a diagonal boundary running from about X1 = -1 at the top to about X1 = 1 near X2 = 0.5, and it narrows toward the top. Roughly half of the Class 3 observations lie outside this region, mostly in the Class 2 region, and new observations would likely behave similarly. Probabilities within the region are again high.